20 Solve algebraically the simultaneous equations

$$x^2 + y^2 = 25$$
$$y - 3x = 13$$

$y = 3x + 13$

Substitute: $x^2 + (3x + 13)^2 = 25$

1 mark
Substitute to eliminate y

×	$3x$	$+13$
$3x$	$9x^2$	$+39x$
$+13$	$+39x$	$+169$

1 mark
Expand the squared bracket

$x^2 + 9x^2 + 78x + 169 = 25$

$-25 \qquad -25$

$10x^2 + 78x + 144 = 0$

$\div 2 \qquad \div 2$

$5x^2 + 39x + 72 = 0$

1 mark
Form a quadratic ready to solve

$5 \times 72 = 360$; $15 \times 24 = 360$ and $15 + 24 = 39$.

×	$5x$	$+24$
x	$5x^2$	$+24x$
$+3$	$+15x$	$+72$

1 mark
Factorise the quadratic

$(5x + 24)(x + 3) = 0$

$5x + 24 = 0$

$-24 \qquad -24$

$5x = -24$

$\div 5 \qquad \div 5$

$x = \dfrac{-24}{5} = -4.8$

$y = 3(-4.8) + 13 = -1.4$

$x + 3 = 0$

$-3 \qquad -3$

$x = -3$

$y = 3(-3) + 13 = 4$

1 mark
Both pairs correctly matched

$x = -4.8, y = -1.4$ or $x = -3, y = 4$

(Total for Question 20 is 5 marks)